Ally Yeseo Shin
Professor A. Yatchew
ECO414: Energy and Regulation
Jun 10, 2025

Sankey Validation and Policy Reflection:
Nunavut's Energy Dependency and Policy Blind Spots

This project analyzes Nunavut's energy flow in a Sankey diagram developed for the year 2021. Comparing quantitatively to official data sources, this report evaluates the accuracy and consistency of the diagram, explores any discrepancies, and reflects on their economic implications. In addition, the assignment examines how current and impending energy policy has the potential to influence Nunavut's unique energy system, and how visual representations like Sankey diagrams may be employed to convey complex energy systems to both policymakers and the populace.

Part A

<table>
<tr><th colspan="2">Type</th><th>My Diagram (PJ, 2021)</th><th colspan="4">Verified source</th><th rowspan="2">Consistent?</th><th rowspan="2">Notes</th></tr>
<tr><th>#</th><th>Type</th><th>#</th><th>#</th><th>Unit</th><th>Year</th><th>Source</th></tr>
<tr><td rowspan="9">1. Energy Production</td><td>Total Primary Production</td><td>0</td><td>0</td><td>-</td><td>-</td><td rowspan="9">Quilliq Energy Corporation, CER</td><td>Y</td><td rowspan="9">No local production; almost all energy is imported (QEC).</td></tr>
<tr><td>Wind</td><td>0</td><td>0</td><td>-</td><td>-</td><td>Y</td></tr>
<tr><td>Nuclear</td><td>0</td><td>0</td><td>-</td><td>-</td><td>Y</td></tr>
<tr><td>Hydro</td><td>0</td><td>0</td><td>-</td><td>-</td><td>Y</td></tr>
<tr><td>Geothermal</td><td>0</td><td>0</td><td>-</td><td>-</td><td>Y</td></tr>
<tr><td>Natural Gas</td><td>0</td><td>0</td><td>-</td><td>-</td><td>Y</td></tr>
<tr><td>Coal</td><td>0</td><td>0</td><td>-</td><td>-</td><td>Y</td></tr>
<tr><td>Biomass</td><td>0</td><td>0</td><td>-</td><td>-</td><td>Y</td></tr>
<tr><td>Petroleum</td><td>0</td><td>0</td><td>-</td><td>-</td><td>Y</td></tr>
</table>

	Solar	0	0.56	GWh	2021	[CER]	Y	Minimal pilot solar installations (e.g., Kugluktuk); production negligible. (CER)
2. Final Energy use by sector	Residental	0	0.273	PJ	2020	[CER]	Y	Value rounds to 0 PJ in the diagram; matches CER data when rounded.
	Commercial	0	0.468	PJ	2020	[CER]	Y	
	Industrial	1.6	1.365	PJ	2020	[CER]	N	
	Transportation	0.5	1.833	PJ	2020	[CER]	N	
3. Energy imports/ exports	Imported Petroleum	6	N/A	N/A	N/A	[CER]	Y	Specific Value is not written but stated: "Almost all of Nunavut's electricity is generated from diesel fuel imported …" (CER)
	Serviced Petroleum	2.1	3	PJ	2020	[CER]	N	
4. Electricity Usage	Electricity	1	0.28	TWh	2021	[CER]	Y	0.28 TWh = 1.008 PJ, so consistent

Part B

1. Final Energy use by sector mismatch

The transportation sector is the most visible discrepancy. In my Sankey diagram, the energy used for transportation was 0.5 PJ, but in the CER 2020 data, the difference was almost three times at 1.833 PJ. This gap is more than just an error.

This discrepancy can be explained by several reasons. First, the base data I used was from 2021, and the CER data is from 2020, so there may be a difference due to the time difference. In particular, since there have been changes in remote work and logistics patterns since COVID-19, the energy of the transportation sector can also vary greatly between years.

In addition, the transportation figure at CER is a separate value of only transportation fuel among all refined petroleum products, and in my diagram, the ratio of total petroleum energy was estimated and divided, so the difference in classification criteria may have had an effect. The problem is likely due to the aggregation vs. disaggregation approach.

Another discrepancy is in industry, where my Sankey diagram shows 1.6 PJ energy consumption, yet just 1.365 PJ based on the CER 2020 figures. The numerical discrepancy appears smaller than in transport, but it is still well over a 10% discrepancy. The likely explanation is the data year discrepancy: my diagram uses 2021 estimates, while CER statistics include 2020 figures. Considering that industrial activity within Nunavut will fluctuate because of seasonal and project-specific operations, i.e., cycles of construction or mining operations, even a one-year difference can cause quantifiable variation. This underlines how time frames must be synchronized when comparing energy models against official data sets.

2. Oil Energy Imports and Service Energy Inconsistency

There is no official figure for the imported petroleum, so instead, I compared the actual amount of petroleum consumed, i.e., "Served petroleum," to CER data on an end-use basis. In my Sankey diagram, the service energy provided as petroleum was 2.1 PJ, and according to CER's 2020 report, 3.0 PJ of defined petroleum products (RPPs) were used.

There is a difference of about 0.9 PJ, or 43%, which is a fairly large discrepancy. One of the main reasons for this discrepancy may be the difference in the way data is interpreted. This is because the "end-use demand" in CER does not include heat loss and transformation loss, whereas the Sankey I worked on is different in that only service energy is calculated separately, considering the overall energy flow.

Additionally, the fact that CER was based on 2020 data and my Sankey was based on 2021 may have made a difference. In particular, oil use may have been lower than usual in 2020 due to a contraction in economic activity caused by COVID-19, which is rather underestimated in the process of estimating 2021 based on these figures.

3. Economic implications of data mismatch

Devaluing or overestimating transportation or oil energy consumption can lead to a misunderstanding of the economic structure. For example, if transportation energy is estimated to be low, the need for infrastructure investment in the sector may be underestimated, even though it is actually a region highly dependent on transportation.

In addition, reflecting less than actual petroleum usage can misidentify how much Nunavut relies on fuel imports, which can give wrong signals to energy security-related or subsidy policies. When discussing future renewable energy transitions, incorrect assumptions about the dependence on existing fossil fuels may be included in the discussion.

Conclusion

My Sankey diagram corresponds to CER data in structural terms, generally, although discrepancies between years and modelling assumptions introduce some significant numerical differences—most visibly in transport and industry sectors. The lack of direct CER data for imports required the use of proxy service energy data, which further uncovered differences in how energy is classified or interpreted between years. While not fatal to the analysis, these inconsistencies highlight the importance of transparency and consistency of data, especially when using visualizations like Sankey diagrams to inform energy policy or investment choices.

(670 words)

Part C

Based on the analysis of the energy flow and policy environment of Nunavut in 2021, it was determined that the following two parts of Assignment 1's guidelines needed improvement. The first is the treatment method for energy flows that are visually omitted from the Sankey diagram, and the second is the application method of the generalized energy loss rate that does not reflect regional differences. The proposal below is intended to reinforce the reflection of clarity and regional specificity in the charting process.

1. Clarifying Total Domestic Use and Service Energy

Original Instruction (from Assignment 1 – Page 4):
"To calculate 'Total Domestic Use' for each primary energy source, sum over 'To Electricity,' 'To Residential,' 'To Commercial,' 'To Industrial,' 'To Non-Energy,' and 'Transportation.'"

Proposed Revision:
"To calculate 'Total Domestic Use' for every major energy source, sum over 'To Electricity,' 'To Residential,' 'To Commercial,' 'To Industrial,' 'To Non-Energy,' and 'Transportation.' Notice that this sum contains both 'service' and 'rejected' energy, while most external data sources (e.g., CER, NRCan) provide only 'service energy' as final end-use demand. In your subsequent validation of the diagram, make a clear distinction between gross flows and net service energy to prevent misinterpretation."

As explained in the essay, hybrid diesel-solar systems or pilot solar installations exist in the Nunavut, but because they are too small, they are marked with 0 PJ on the Sankey diagram and are completely omitted visually. This may lead to a misunderstanding as if there is no renewable energy conversion effort. Therefore, even if the microscopic flow is visually omitted, guidelines that specify the existence of the project through annotation or legend are needed.

2. Adjusting Rejection Rate Assumptions for Regional Accuracy

Original Instruction (Assignment 1 – Page 5):
"Finally, it is necessary to calculate the proportion of Energy from each final use category that is rejected. Rejection rates are as follows... Residential: 35%, Commercial: 35%, Industrial: 20%, Transportation: 75%."

Proposed Revision:
"Finally, calculate the proportion of Energy from each end use category that is rejected, using the rates given below. Note, however, that these assumptions are based on generalized estimates and might not reflect regionally specific circumstances. For example, energy consumption for transportation in northern territories like Nunavut can vary from year to year depending on climate, resupply logistics, and infrastructure factors. When comparing against official statistics, state clearly whether you are using modelled service energy or reported end-use data."

Nunavut may have a greater energy loss rate than other regions due to factors such as extreme climate, inefficient housing and heating systems, and distributed microgrids. In particular, there is a possibility that energy loss in the transportation and housing sectors is higher than the national average. Nevertheless, Assignment 1 applies generalized values at once, so there is a limit to which it cannot reflect regional specificity. To compensate for this, the possibility of regional adjustments should be included in the guidelines.

Part D

1. Extreme Dependence and Economic Vulnerability

The Nunavut Territory's energy flow in 2021 is simple, but it clearly reveals the region's economic structure, energy dependence, and policy weakness. The Sankey diagram shows that all of Nunavut's energy is based on diesel imports. In other words, there is no energy self-sufficiency in the region, and the entire region is dependent on a single fuel source.

This trend shows that Nunavut's economic structure is tied to an extremely simplified supply chain. It has the largest area (about 2 million km^2) in Canada, but has a population of only about 40,000 people, and a population density of 0.02 people/km^2, which is globally rare. These topographical and climatic conditions, where infrastructure is impossible, have made Nunavut a trade-vulnerable region that relies absolutely on external supply. Not all communities are connected by electricity grids or gas pipes, and only one year's worth of diesel is imported and stored through barge transportation or flights in summer. In the event of a supply disruption, there are no alternative means, which goes beyond simple inconveniences and leads to energy security threats.

Economically, Nunavut is an ultra-high-priced area due to transportation costs. Most of the daily necessities and energy resources are transported by plane, so food and fuel prices are significantly higher than those of southern Canada. To alleviate this, the federal government is subsidizing some foods through the Nutrition North program, but even this does not apply to most processed foods or non-necessities, so there is a lot of controversy over its effectiveness. Such a structure goes beyond a simple increase in the cost of living, leading to a surge in the cost of survival directly related to energy use. In other words, it can be seen as an area where the concept of energy poverty is actually working.

2. Policy Blind Spots and Future Possibilities

It is somewhat disappointing to see how the current energy policy is affecting this environment. Unlike other parts of Canada, Nunavut is not subject to a carbon tax. This may have been considered to avoid controversy over equity in responding to climate change, but it is actually creating a policy blind spot. In particular, Nunavut is one of the regions with the highest per capita greenhouse gas emissions in Canada as a whole, due to the demand for heating, due to the extreme climate and lagging facility efficiency. Nevertheless, so far, few approaches have been made to fundamentally solve this problem at the policy level.

There is certainly a movement to transform the energy structure of Nunavut. Small-scale solar installations or hybrid diesel-solar projects are being carried out on a trial basis in some communities.

However, this is also insignificant enough to appear as 0.0 PJ on the Sankey diagram. It remains in the "pilot" or "basic research stage" rather than the structural transition, and it is still a long way from making actual flow changes. This is due to a combination of high cost, low technology access, and, above all, the fact that renewable energy does not guarantee as much reliability as diesel in the harsh climate of the Arctic Circle.

However, the Nunavut Lands and Resources Devolution Agreement signed in 2024 can be a significant inflection point. The transfer of ownership of the resources and land held by the federal government to the Nunavut Territory opens up the possibility of the region exploring and utilizing its own energy resources. Although it will not change much in the short term, it can serve as the basis for community-led renewable energy transition and sustainable energy policy establishment in the mid to long term.

3. The Role of Sankey Diagrams in Understanding and Decision-Making

Finally, visualization tools like Sankey diagrams play an excellent role in the simple and intuitive representation of these complex and overlapping problems. The visual expression that all flows in one direction, particularly in systems with a simple structure and little change, such as Nunavut, has the effect of emphasizing the lack of energy diversity and dependence. This tool also enables policymakers and the general public to understand the nature of the problem and intuitively recognize the need for change.

Of course, this tool has limitations in omitting some flows or excluding qualitative elements, but in extreme cases such as Nunavut, such simplification works convincingly. For example, sectors expressed as "0 PJ" in the energy flow may actually exist microscopically, but the method of being completely omitted visually becomes a strong message that emphasizes the system's deficiencies.

In conclusion, the energy flow of Nunavut is simple, but that is the essence of the problem. Only one fuel, only one supply path, only one consumption pattern. This is not sustainable in terms of policy, economy, and environment, and practical policy intervention for structural change is needed before it is too late. The Sankey diagram is a powerful tool to show all these problems at a glance, and it is time for practice beyond its visualization to follow.

(830 words)

References

Government of Canada, Canada Energy Regulator. (2024, September 10). *CER – Provincial and Territorial Energy Profiles – Nunavut*. https://www.cer-rec.gc.ca/en/data-analysis/energy-markets/provincial-territorial-energy-profiles/provincial-territorial-energy-profiles-nunavut.html

Power in Nunavut | Qulliq Energy Corporation. (n.d.). https://www.qec.nu.ca/power-nunavut